INCH–POUND

MIL–PRF–51191G(EA)
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SUPERSEDING
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w/AMENDMENT 1
23 August 2010

PERFORMANCE SPECIFICATION

HEATER, AIR, ELECTRICAL, FILTER UNIT, M3

Reactivated for new design after 12 June 1997

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric air heaters that provide for heating air supplied from Gas–Particulate Filter Units (GPFUs) to military personnel who wear Nuclear–Biological–Chemical (NBC) protective masks during cold weather operations.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: RDCB–DEA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to APGR–EcbcSpecsTeam@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MS27142	–	Connector, Plug, Electric− Pin Contact, No. 12, 14, 16 AWG, Waterproof
MS27144	–	Connector, Plug, Electrical− Socket Contact, No. 14 and 16 AWG, Waterproof
MIL−STD−461	–	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL−STD−810	–	Environmental Engineering Considerations and Laboratory Tests
MIL−STD−1168	–	Ammunition Lot Numbering and Ammunition Data Card

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

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DRAWINGS

U.S. ARMY EDGEWOOD CHEMICAL BIOLOGICAL CENTER

5−19−1782 − Heater, Air, Electric, Filter Unit, M3

(Copies are available from U.S. Army Edgewood Chemical Biological Center, ATTN: RDCB−DEA−D, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010−5424.)

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

12910415 − Guard Assembly

(Copies of this drawing may be requested online at pica.drawing.request@conus.army.mil or from U.S. Army Armament, Research, Development and Engineering Center, ATTN: RDAR−EIS−PE, Picatinny Arsenal, NJ 07806−5000.)

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 − Unified Inch Screw Threads
ASME B1.2 − Gages and Gaging for Unified Inch Screw Threads
ASME B1.3 − Screw Thread Gaging Systems for Acceptability: Inch and Metric Screw Threads

(Copies are available online at http://www.asme.org/Publications/ or ASME, P.O. Box 2300, Fairfield, NJ 07007−2300.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this

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document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Interface requirements.

3.2.1 Maximum critical dimensions. The size of the air heater shall not exceed the following critical dimensions: The maximum length (distance between furthest inlet port location and furthest outlet port location) shall be no greater than 7.32 inches. The distance between the bottom of the mounting bracket (base plate) and the highest point on the controller housing (excluding the control knob) shall be no greater than 5.95 inches when the center axis of the controller is oriented perpendicular to the mounting bracket. The controller shall not exceed a "footprint" of 2.78 x 3.70 inches, whereby the 2.78 inches is parallel to the longitudinal axis of the heater to allow necessary clearance for hose connections. A "footprint" is defined as the maximum length and width of the controller in any plane parallel to the heater base when the controller is upright relative to the base plate.

3.2.2 Compatibility with interfacing parts. In addition to the requirements in 3.2.1, the following requirements shall be met to insure compatibility of the air heater with the vehicle and its Gas Particulate Filter Unit (GPFU) system. The degree of movement, orientation and number of the air inlet and outlet ports shall be equivalent to those on Drawing 5−19−1782 in form, fit and function. Two threaded plugs, which are form, fit, and function equivalent to those on Drawing 5−19−1782, shall be installed in the side inlet and outlet ports. Material selection of the plugs and heater's housing shall prevent galvanic corrosion between the plugs and the housing. (Additional plugs may be required by solicitation or special packaging instruction.) Female screw threads in the inlet and outlet ports of the heater and those in the heater for the attachment of the controller to the heater shall conform to ASME B1.1 and be compatible in form, fit, and function to threads defined for the inlet and outlet ports and in Detail B of Drawing 5−19−1782, respectively. The cable connectors on the air heater shall be interoperable and interchangable with those shown on Drawing 5−19−1782 and in MS27142 and MS27144. The female and male connectors shall remain firmly attached except when normal hand force is used to separate them. The hole size and pattern in the mounting bracket of the heater shall conform to Detail C of Drawing 5−19−1782. The heater with controller shall have the capability to be mounted in the guard assembly (part no. 12910415) without interference.

3.2.3 Controller features. The controller shall be capable of being removed from the heater and remotely mounted inside the vehicle on a mounting that has the same threaded hole pattern as is described by Detail B of Drawing 5−19−1782. The controller shall enable the operator to change the air stream temperature over a continuous range via a temperature

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control knob. When the control knob is turned fully counterclockwise, no heat shall be supplied to the air stream. The controller assembly shall be totally enclosed and watertight with all electrical control circuitry (relays, switches, solid state electronics, etc.) sealed inside. The center line of the controller shall be capable of being moved at least 45 degrees in either direction (clockwise or counter−clockwise) to the vertical plane which runs through the longitudinal axis and is perpendicular to the mounting bracket.

3.3 Operating requirements.

3.3.1 Compatibility with vehicle electrical system. The heater shall be capable of withstanding normal vehicular voltage surges and spikes such as the following. The heater with controller shall be capable of withstanding a surge voltage of 39 volts for 10 seconds and recover within 0.015 second. The heater with controller shall meet all requirements in 3.3 after withstanding a spike voltage of 250 volts for 0.015 milliseconds.

3.3.2 Operability during movement. The heater shall meet all operating requirements in 3.3 when the air inlet and outlet ports are rotated 180 degrees to each side of the vertical position. The heater shall not malfunction due to opening or shorting of internal wiring.

3.3.3 Heating capability and power consumption. The heater shall be capable of receiving 4.5 cubic feet per minute (cfm) of air, minimum, (when corrected to standard conditions of 70°F and 1 atm) at its inlet at a temperature of 0 ± 4°F and discharging this air at its outlet at a minimum temperature of 120°F within 15 minutes after starting heater set to its maximum output level. Heater shall not consume more than 250 watts of power.

3.3.4 Resistance to airflow. With the heater at ambient room temperature, the resistance to airflow shall not exceed a differential pressure of 0.10 inches of water.

3.3.5 Operability with thermostatic control. The heater shall operate in a room temperature environment of 70°F and be capable of producing a regulated outlet air temperature of 120±4°F for input voltage of 24 to 30 Volts direct current (Vdc) and 115.5±8.5°F for input voltage of 18 Vdc within 10 minutes of operation when the inlet air flow rate is 4.5 cfm. The heater shall maintain those outlet air temperatures for a period of not less than 5 minutes without adjustment of the controller. During these conditions, the case temperature shall not exceed 126°F.

3.3.6 Leakage resistance. Air leakage shall be not greater than 1.00 cubic inch (16.4 cubic centimeters) per minute when the filter is internally pressurized to at least 12 inches (30.5 centimeters) of water. Teflon™ or polytetrafluoroethylene tape shall be applied to the threads of the plugs if this prevents sticking or leakage around plugs.

3.3.7 Electromagnetic emissions. The heater shall not produce electromagnetic emissions of such magnitude that mission essential communications equipment and electronic systems are adversely effected. Measured electromagnetic emission levels shall be within the limits

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specifed in MIL−STD−461 for conducted emissions from power leads and radiated electric field emissions.

3.4 Environmental requirements.

3.4.1 Hot/cold temperature and plug duty cycles.

3.4.1.1 Hot temperature survivability. The heater shall survive hot temperature storage temperatures.

3.4.1.2 Cold temperature survivability. The heater shall survive cold temperature storage temperatures.

3.4.1.3 Leakage resistance after plug duty cycles. After the plugs are installed and removed 50 times in a cold environment and 50 times in a hot environment, the plugs shall not exhibit damage and the heater shall meet operability with thermostatic control (3.3.5) and leakage resistance (3.3.6) requirements.

3.4.1.4 Cold temperature operation. When the heater is operating in a temperature environment of −45°F, it shall be capable of producing a regulated outlet air temperature of 70±4°F within 15 minutes of operation and shall maintain this temperature for a period of not less than 5 minutes without adjustment of the controller.

3.4.2 High humidity. The heater shall survive hot humid storage temperatures. Following humidity exposure, the heater shall meet operability with thermostatic control (3.3.5) and leakage resistance (3.3.6) requirements.

3.4.3 Salt fog. After the heater's inlet and outlet port plugs are exposed to salt fog environment for 96 hours and when the plugs are installed on the inlet and outlet ports, the heater shall meet operability with thermostatic control (3.3.5) and leakage resistance (3.3.6) requirements.

3.4.4 Mechanical shock. The heater shall survive shock levels expected during field use. Following shock waves, the heater shall meet operability with thermostatic control (3.3.5) and leakage resistance (3.3.6) requirements.

3.4.5 Vibration. The heater shall survive vibration levels expected during field use. Following vibration, the heater shall meet operability with thermostatic control (3.3.5) and leakage resistance (3.3.6) requirements.

3.5 Support and ownership requirements.

3.5.1 Minimum operating life. The heater shall maintain the outlet temperature of 120±4°F for a continuous period of at least 250 hours.

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3.5.2 Identification and markings. Each heater shall contain the following indentification information, clearly and legibly marked: “Heater, air, electric, filter unit, M3”; 27.5 Vdc rating, nominal maximum power rating in watts at the specified 27.5 Vdc input voltage; lot number, model and/or serial number, and contract number. Electrical leads between the controller and heater shall be marked with permanent markings on both sides of the electrical connectors to ensure that leads are identifiable for correct reconnection. The inlet and outlet airflow ports shall be clearly and legibly marked. When the heater is not energized, the control knob shall clearly indicate “off.” The direction of turn for warmer air shall be marked on the controller. Clarity of the identification information and markings shall not be degraded by temperature and humidity stress.

3.5.3 Workmanship. The heater shall be free from missing or loose parts; dented, cracked, or chipped surfaces; oil or foreign matter adhering to outside surface; and foreign matter blocking the air ports.

3.5.4 Color and finish. The color of the exterior of the heater shall be white or off−white with gloss finish except for threads, controller knob, electrical connectors and plated parts, and any other area designated on Drawing 5−19−1782. All external metallic surfaces or coatings applied to such surfaces shall resist corrosion and heat stress.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection

4.2.1 Sample. The first article sample shall consist of 20 heaters manufactured using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2). Five heaters are allocated to each of the four sample groups in table I. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspections to be performed. The first article sample heaters shall be inspected and tested in the sequence shown in table I.

4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

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TABLE I. First article inspection

Paragraph Title	Sample Group	Requirement paragraph	Verification paragraph
Interface			
Maximum critical dimensions	I	3.2.1	4.4.1.1
Compatibility with interfacing parts	I	3.2.2	4.4.1.2
Controller features	I	3.2.3	4.4.1.3
Operation			
Compatibility with vehicle electrical system	II	3.3.1	4.4.2.1
Operability during movement	II	3.3.2	4.4.2.2
Heating capability and power consumption	II	3.3.3	4.4.2.3
Resistance to airflow	II	3.3.4	4.4.2.4
Operability with thermostatic control	II	3.3.5	4.4.2.5
Leakage resistance	II	3.3.6	4.4.2.6
Electromagnetic emissions	II	3.3.7	4.4.2.7
Environment			
Hot/cold temperature and plug duty cycles	III	3.4.1	4.4.3.1
Hot temperature survivability	III	3.4.1.1	4.4.3.1(a)
Cold temperature survivability	III	3.4.1.2	4.4.3.1(c)
Leakage resistance after plug duty cycles	III	3.4.1.3	4.4.3.1(b)(d)
Cold temperature operation	III	3.4.1.4	4.4.3.1(e)
High humidity	III	3.4.2	4.4.3.2
Salt fog	III	3.4.3	4.4.3.3
Mechanical shock	III	3.4.4	4.4.3.4
Vibration	III	3.4.5	4.4.3.5
Support and ownership			
Minimum operating life	IV	3.5.1	4.4.4.1
Identification and markings	I	3.5.2	4.4.4.2
Workmanship	I	3.5.3	4.4.4.3
Color and finish	I	3.5.4	4.4.4.4

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the heaters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and at essentially the same time. Each lot shall be identified and controlled in accordance with MIL−STD−1168.

4.3.2 Sampling. Sampling of lots shall be in accordance with classification of characteristics in 4.3.5, and when specified, table II. Samples shall be selected at random.

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4.3.3 Inspection procedure. Sample items shall be examined and tested in accordance with the classification of characteristics in 4.3.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

TABLE II.

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals who use or maintain the product. Characteristics whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item are also critical characteristics. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

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4.3.5 Classification of characteristics. Conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table II for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.5	Heater, Air, Electric, Filter Unit, M3		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Resistance to airflow	Table II, level IX	3.3.4	4.4.2.4
102	Operability with thermostatic control	Table II, level IX	3.3.5	4.4.2.5
103	Leakage resistance	Table II, level IX	3.3.6	4.4.2.6
104	Mounting holes correct	Table II, level IX	3.2.2	CE
105	Thread size of outlet and inlet openings correct	Table II, level IX	3.2.3	CE
106	Electrical lead markers correct	Table II, level IX	3.5.2	VI
107	Inlet and outlet ports marked correct	Table II, level IX	3.5.2	VI
108	Markings for on/off status and temperature control correct	Table II, level IX	3.5.2	VI
109	Workmanship	Table II, level IX	3.5.3	VI
Minor				
201	Other identification marking correct	Table II, level XI	3.5.2	VI
202	Color and finish	Table II, level XI	3.5.4	VI

NOTES:

CE – Commercial inspection equipment
VI – Visual inspection

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4.4 Verification methods and procedures.

4.4.1 Interface requirements verification.

4.4.1.1 Maximum critical dimensions. Verify that critical dimensions do not exceed maximum values specified in 3.2.1 by examintion with measurement.

4.4.1.2 Compatibility with interfacing parts. Verify orientation and number of ports conforms to Drawing 5−19−1782. Verify that the screw threads in the inlet and outlet ports and those in the heater for the attachment of the controller conform to 3.2.2 (Drawing 5−19−1782) by use of appropriate gages which conform to ASME B1.2 and B1.3. Verify no galvanic corrosion between the plugs and housing by materials study or analysis. Verify that all pairs of male and female electrical connectors external to the controller and heater may be pulled loose using ordinary hand force and reattached. Verify hole size and pattern of mounting bracket are as specified in 3.2.2.

4.4.1.3 Controller features. Remove the controller from the heater and reattach it. Move the controller 45 degrees in either direction to the vertical plane that is perpendicular to the mounting bracket. Verify that mounting holes in controller are located consistent with the requirements of 3.2.3.

4.4.2 Operating requirements verification.

4.4.2.1 Compatability with vehicle electrical system. The heater shall be tested with test equipment capable of measuring surge and spike voltages described herein. Operate the heater with the input voltage set to a nominal steady state voltage of 27.5 Vdc. Apply a continuous 39 Vdc surge voltage for 10 seconds to the power input terminals and determine whether the recovery time is less or equal to 0.015 sec. The recovery time is the interval between the time a voltage deviates from the steady state voltage limits (25 to 30 Vdc) and the time it returns and remains within that same range. Then apply a 250 Vdc spike voltage for 0.015 milliseconds.

4.4.2.2 Operability during movement. Rotate inlet and outlet ports 180 degrees to each side of the vertical position while the heater is energized by a 27.5 Vdc source and operating.

4.4.2.3 Heating capability and power consumption. Two calibrated thermocouples shall be installed in the system, one at the inlet and one at the outlet port, to measure the inlet and outlet air temperature. The thermocouples shall be installed approximately one inch from their respective ports. The heater shall be connected to a suitable power source capable of producing 27.5 Vdc. A voltmeter and ammeter shall be connected to the electrical circuit. The heater shall be conditioned for 24 hours at 0±4°F prior to testing. Air, at 0±4°F and flowing at 4.5 cfm, shall pass through a calibrated gas metering device and then through the heater. Two of the 4 heater samples shall be tested with air flowing through the end inlet and outlet ports and the remaining two shall be tested with air flowing through the side inlet and

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outlet ports. Set the input voltage (voltage while operating) to 27.5 Vdc. Rotate the control knob fully clockwise (full on), operate the heater continuously for at least 15 minutes, and record the current consumption during that period. After 15 minutes of operation, determine whether the power consumption, outlet air temperature and case temperature meet the requirements in 3.3.3. (Note: turning the control knob fully clockwise for extended periods may result in outlet air temperatures that could be harmful to the hose and heater finish.)

4.4.2.4 Resistance to airflow. The pressure differential shall be measured with a suitable manometer across the side inlet and outlet ports of the heater when the heater is receiving 4.5 cfm of air at ambient room temperature (70 ± 4°F) and not energized.

4.4.2.5 Operability with thermostatic control. Two calibrated thermocouples shall be installed in the system, one at the side inlet port and one at the side outlet port, to measure the inlet and outlet air temperature. The thermocouples shall be installed within one inch from their respective ports. Six additional calibrated thermocouples shall be attached to six scattered points, at least one of which shall be attached to the outlet boss, on the heater surface. With a voltmeter and ammeter connected to the electrical circuit, set the input voltage (voltage while operating) to 27.5 Vdc . Ambient room air at 70± 4°F, flowing at 4.5 cfm, shall pass through a calibrated metering device and then through the heater. Turn the control knob fully clockwise until outlet air temperature rises to 120°F, then adjust knob until outlet air temperature stabilizes at 120±4°F. Maintain an outlet air temperature of 120 ± 4°F for a period of at least 5 minutes without adjustment of the controller knob. The heater is defective if it requires more than 10 minutes to attain the required output temperature, or if it fails to maintain the output temperature within ± 4°F without adjustment of the controller knob. Compare the maximum measured case surface temperature against the specified value in 3.3.3. Repeat aforementioned test for an input voltage of 18.0 Vdc and for an input voltage of 30.0 Vdc (group II, first article samples only).

4.4.2.6 Leakage resistance. With the side inlet and side outlet ports sealed with plugs, connect the end inlet port to a source of pressurized air with an airflow meter and appropriate regulating valve(s) between the heater and pressurized air source (see Figure 1). Connect a manometer or pressure gage capable of measuring a static pressure of 12.0 inches of water gage (iwg) with an accuracy of ± 0.1 iwg to the end outlet port by means of appropriate adapters. The airflow meter shall be capable of measuring an airflow of 1.00 cubic inch per minute with an accuracy of ± 0.06 cubic inch per minute. Open the airflow and pressure regulating valve(s) and adjust the flow until a static pressure of at least 12 inches of water is established within the heater. The steady state flow rate that is required to maintain that pressure is the air leakage value. The heater is considered defective if the air leakage value exceeds the specified maximum value of 1.00 cubic inch per minute.

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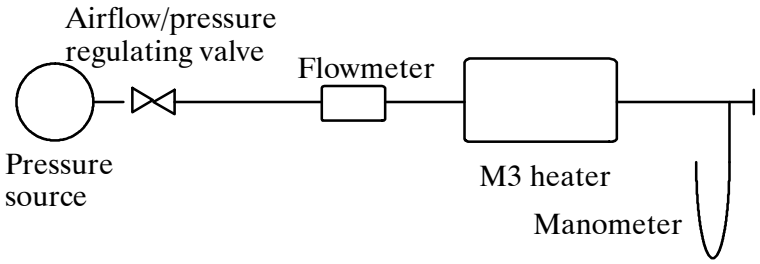


FIGURE 1. Leak test apparatus

4.4.2.7 Electromagnetic emissions. The heater shall, as a minimum, be tested for conducted and radiated emissions in accordance with MIL−STD−461 and Table III at minimum, mid−range and maximum controller settings.

TABLE III. Test methods for electromagnetic emissions

MIL−STD−461 Test Method	Test Description	Acceptance criteria
CE102	Conducted emissions, power leads, 10 kHz − 10 MHz	Basic curve limits (dBμV) in Figure CE102−1
RE102	Radiated emissions, electric field, 10 kHz to 18 GHz	Navy Mobile and Army limits (dBμV/m) in Figure RE102−4

4.4.3 Environmental requirements verification.

4.4.3.1 Hot/cold temperature and plug duty cycles. While the heater is not in operation, subject it to the following sequence of temperature exposures.

(a) Place the heater in a high temperature chamber, and raise the chamber temperature to 160°F. Maintain for a period of not less than 4 hours or until the test item stabilizes.

(b) Remove the heater from the hot temperature chamber and examine for damage. Within 5 minutes of removal from chamber, remove and install inlet and outlet port plugs 50 times. Examine plug threads for damage. Then conduct tests for operability with thermostatic control (4.4.2.5) and leakage resistance (4.4.2.6).

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(c) Transfer the heater, within 5 minutes, to a cold chamber with an internal temperature of −60°F. Maintain for a period of not less than 4 hours or until the test item stabilizes.

(d) Remove the heater from the low temperature chamber and examine for damage. Within 5 minutes of removal from chamber, remove and install inlet and outlet port plugs 50 times. Examine plug threads for damage. Then conduct tests for operability with thermostatic control (4.4.2.5) and leakage resistance (4.4.2.6).

(e) Test the heater for cold temperature operation as follows: Place the heater in a chamber and condition it at −45° F for four hours. Operate the heater in this environment with the input voltage set to 27.5 Vdc and the airflow set to 4.5 cfm, also at −45°F. Turn the control knob to the full "on" position and monitor the outlet air temperature until it attains 70 °F or until 15 minutes have elapsed. If the outlet air temperature rises to 70°F within 15 minutes, set the control knob to maintain this temperature within ± 4°F. Maintain an outlet temperature of 70 ± 4°F for a period of at least 5 minutes without adjustment of the controller knob. The heater is defective if it requires more than 15 minutes to attain the 70°F output temperature, or if it fails to maintain the output temperature within ± 4°F without adjustment of the controller knob.

4.4.3.2 Humidity. While the heater is not operating, subject heater to ten 24−hour temperature−humidity cycles in accordance with MIL−STD−810, Method 507.5, Procedure II − Aggravated Cycle (Figure 507.5−7). Following the ten 24−hour temperature−humidity cycles, examine heater for evidence of damage and degradation to markings. Then conduct tests for operability with thermostatic control (4.4.2.5) and leakage resistance (4.4.2.6).

4.4.3.3 Salt fog. The heater shall be be subjected to salt fog testing in accordance with Method 509.5 of MIL−STD−810. Test duration shall consist of four 24−hour periods of alternating salt fog challenge and drying periods. At the end of the salt fog challenge and drying cycles, remove salt deposits with distilled water and dry the heater at ambient conditions for at least 48 hours. Examine heater and its plugs for evidence of corrosion. Conduct follow−on testing for operability with thermostatic control (4.4.2.5) and leakage (4.4.2.6).

4.4.3.4 Mechanical shock. While the heater is "off", the heater shall be subjected to shock testing in accordance with MIL−STD−810, Method 516.6, Procedure I − Functional Shock. Three half sine wave shock pulses shall be applied in positive and negative directions along each of three mutually perpendicular axes (vertical, longitudinal, transverse) for a total of 18 total shock pulses. Pulse amplitude, cross−over frequency, and duration shall be in accordance with Table 516.6−I of MIL−STD−810. After shock test, examine heater for damage. Then conduct tests for operability with thermostatic control (4.4.2.5) and leakage resistance (4.4.2.6).

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4.4.3.5 Vibration. While the heater is "off, conduct a vibration test in accordance with MIL−STD−810, Method 514.6, Procedure I − General Vibration. Use the vibration profile in Table 514.6C−VI− Category 4 − Composite wheeled vehicle vibration exposure. Vibration duration shall be 2 hours for each of the three orthogonal axes. After vibration in each axis, examine heater for damage. (An alternate vibration procedure conforming to revision F of MIL−PRF−51191 may be used if approved by the Government.) Following vibration, conduct tests for operability with thermostatic control (4.4.2.5) and leakage resistance (4.4.2.6).

4.4.4 Support and ownership requirements verification.

4.4.4.1 Minimum operating life. A voltmeter and ammeter shall be connected to the electrical circuit. Ambient room temperature air shall be passed through a calibrated metering device through the heater, and exhausted outside the test area so that the heated exhaust will have no influence on the ambient room temperature. Input voltage shall be 27.5 Vdc. At the start of the test, while the heater is operating at an airflow rate of 4.5 cfm , the controller shall be adjusted to an output air temperature of 120 ± 4°F. Aforementioned temperature shall be maintained for a minimum of 250 hours.

4.4.4.2 Identification and markings. Verify requirements by examination.

4.4.4.3 Workmanship. Verify requirements by examination.

4.4.4.4 Color and finish. Verify requirements by examination.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity. Note: except for interplant shipment, air heaters will normally be shipped with caps or covers which protect all inlet and outlet ports from foreign material.

6. NOTES

6.1 Intended use. The air heater covered by this specification is a component of the NBC GPFU systems utilized in various military vehicles, and is used to warm respiratory air sup-

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plied to individual protective masks and headpieces. Air from inside the vehicle is blown into a precleaner which contains a particulate filter and two gas filters before entering the heater.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Packaging requirements (see 5.1).
(c) First article:
  (1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
  (2) Name and address of test facility and shipping instructions when testing is performed by the Government.
  (3) Time required for the Government to notify the contractor whether or not to proceed with production.
(d) Responsibility of inspection.

6.3 Safety. It is desired that the heater design comply with the safety requirements of UL 499 for internal wiring, electrical insulation, thermal insulation, and insulation resistance.

6.4 Subject term (key word) listing.

Filtered air
Gas Particulate Filter Unit (GPFU)
Respiratory air
Respiratory mask

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Tank air heater
Temperature controller

6.5 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:

Army − EA

Project No. 4240−2011−001

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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